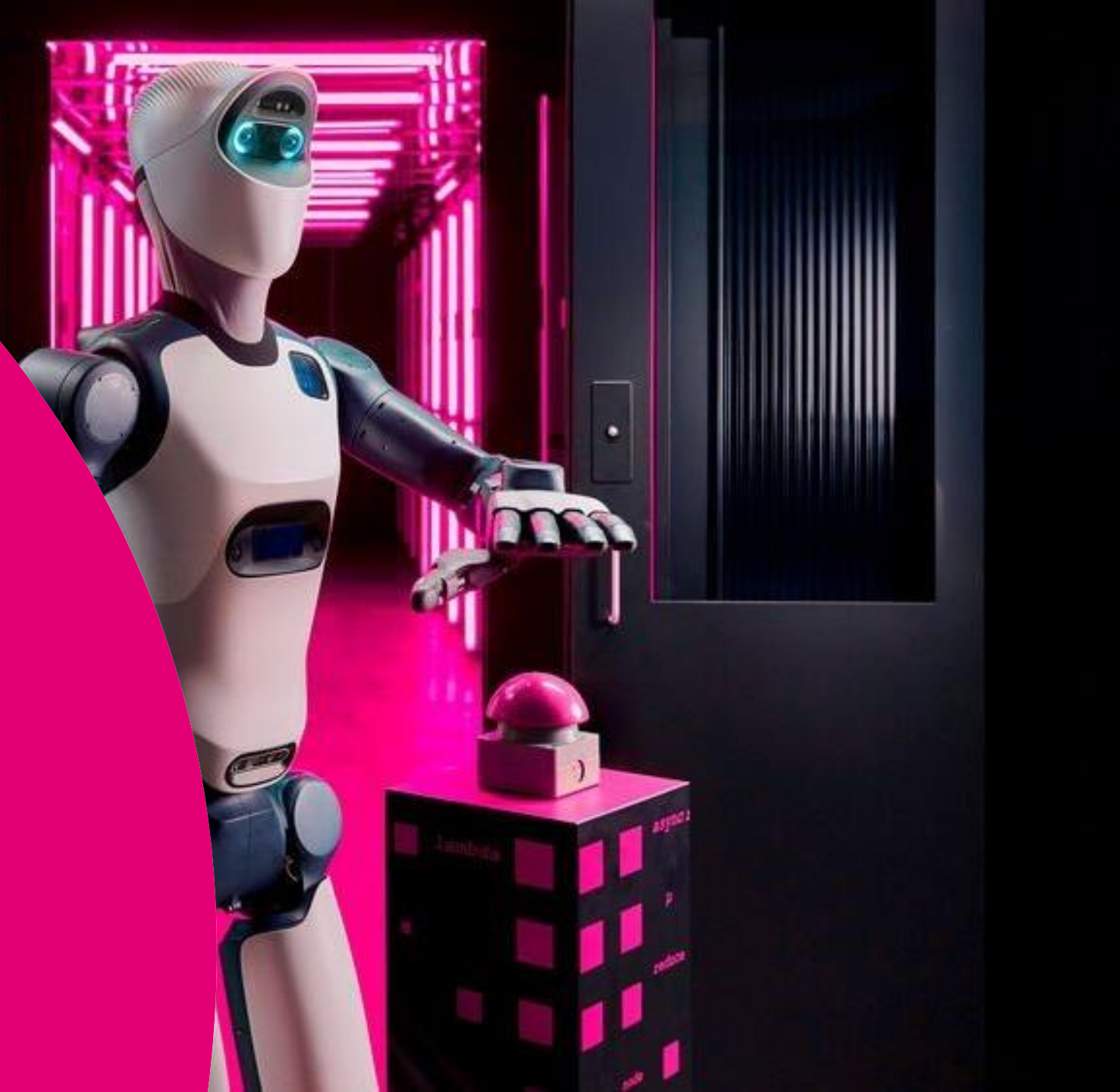
DT Use Cases for Agentic AI: Multi-Agent Intelligence for Service Operations & Agent Discovery

Roland Schott et al. | Shenzhen | March 19th 2026



01

Unexpected Incidents Management with Multi-Agent Collaboration

Network Experience Improvement with Agentic AI

Use Case Examples for Unexpected Incidents Management with Multi-Agent Collaboration:

1. Proven RAN Guardian Agent live in Germany reducing time to manage major events from hours to around a minute – with over 100 remediation actions autonomously triggered in the first month after launch.

2. Ambition for a Service Oriented AI System that predicts, detects, and resolves network issues before customer are impacted.

3. Ambition:
   MINDR – Multi-Agentic Intelligent Network Diagnostics & Remediation, a multi-agentic artificial intelligent (AI) system to enable autonomous diagnostics and operations across complex, multi-domain telecommunication networks in partnership with external partners using agent-to-agent protocol.

RAN Guardian Agent: Unexpected Incidents Scenarios and Challenges

Scenarios

Traffic Burst and Performance Degradation caused by Large number of User Access

Marathons

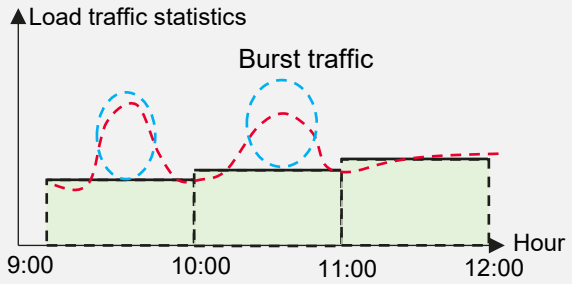
Traffic Jams

Sport Events

	Scenario1 Un-planned Events	Scenario2 Planned Events
Typical Event	Accidents/Traffic Jams/...	Concerts/Sporting/...
Characteristics	Unfixed time & location	Fixed time & location
Suggestion	Focus on Downlink throughput optimization	Focus on Uplink throughput optimization

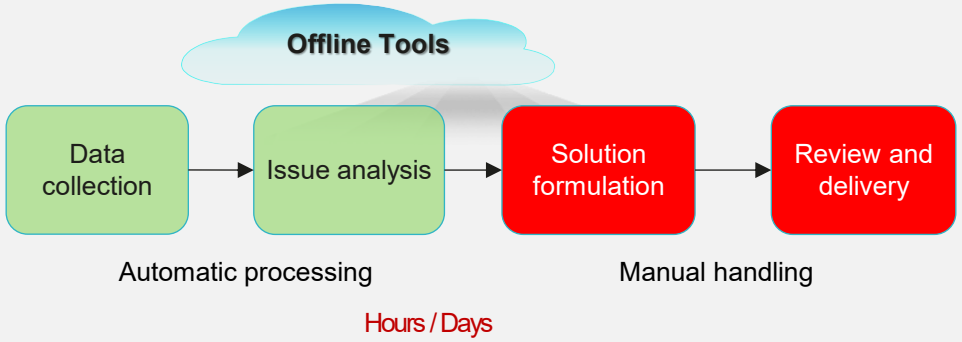
Challenges

Challenge 1: Perception and Prediction of Traffic Burst



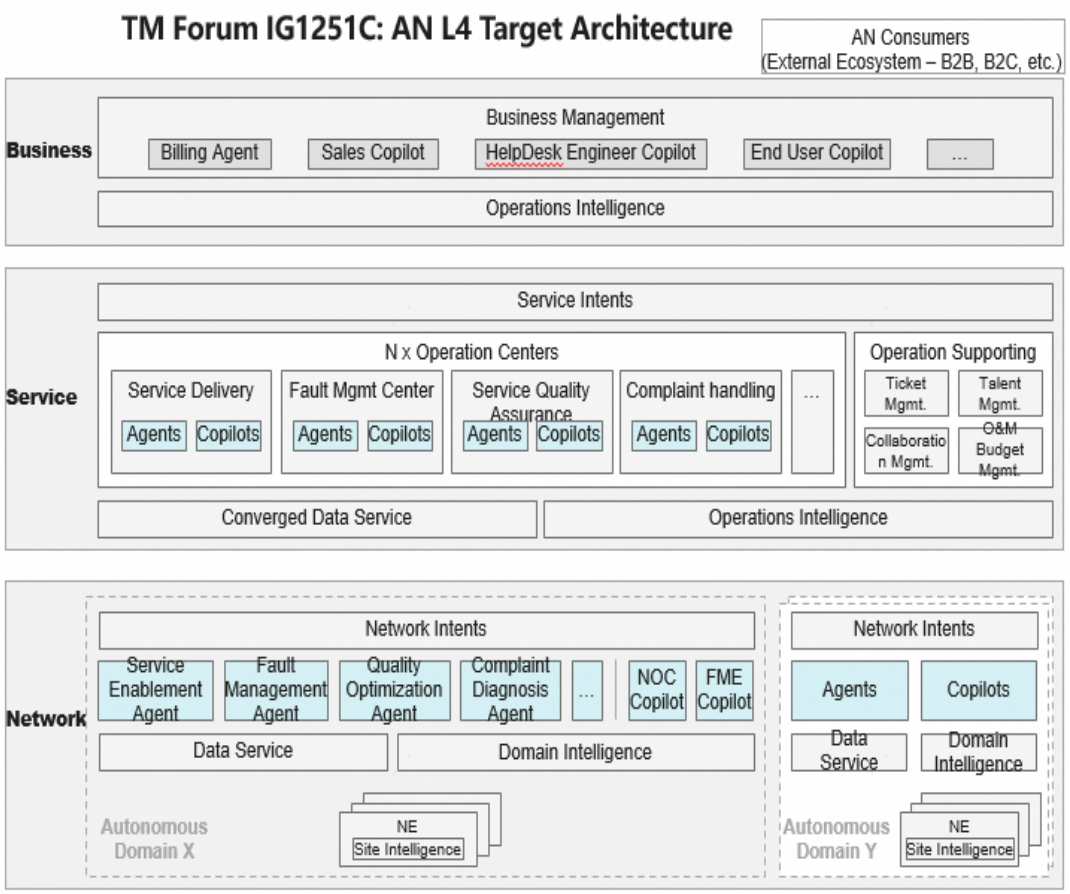
Hour-level traffic monitoring cannot capture the traffic burst occur at minute-level

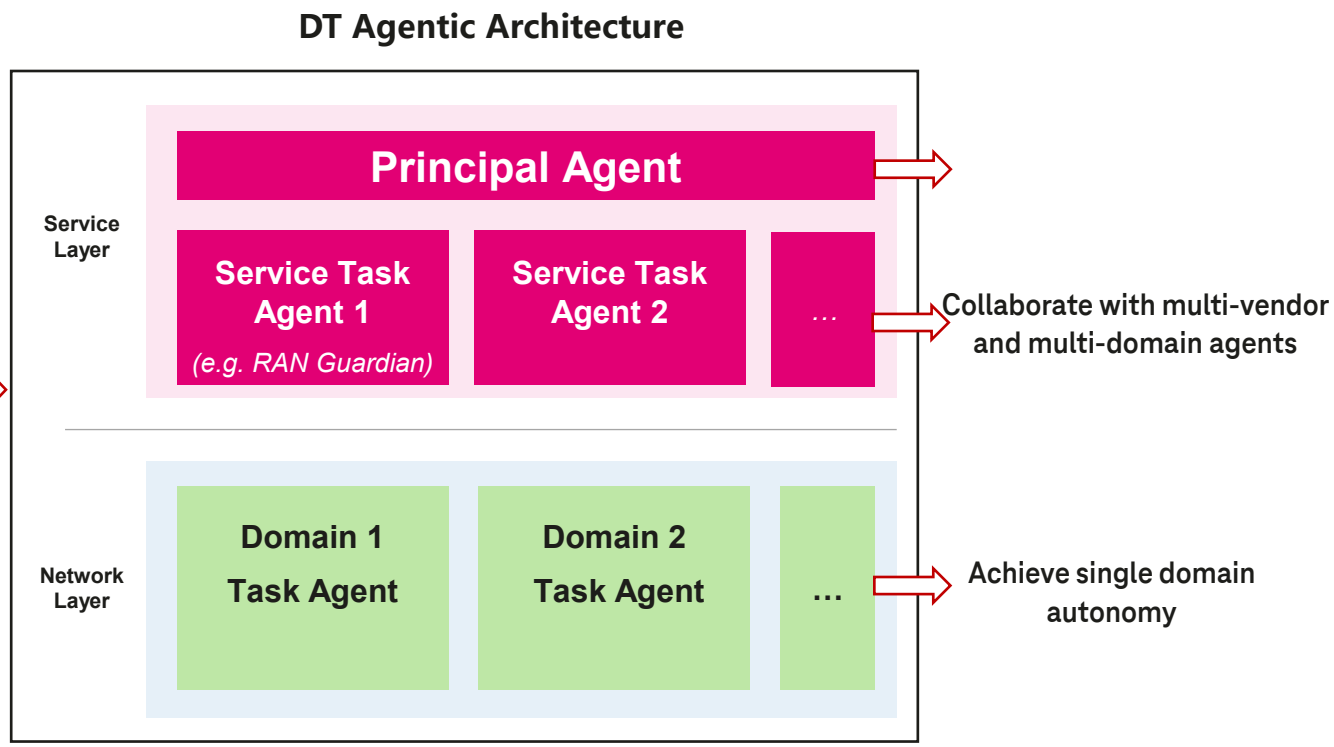
Challenge 2: Manual Optimization is Low-efficiency for Experience Assurance/Compliant Handling



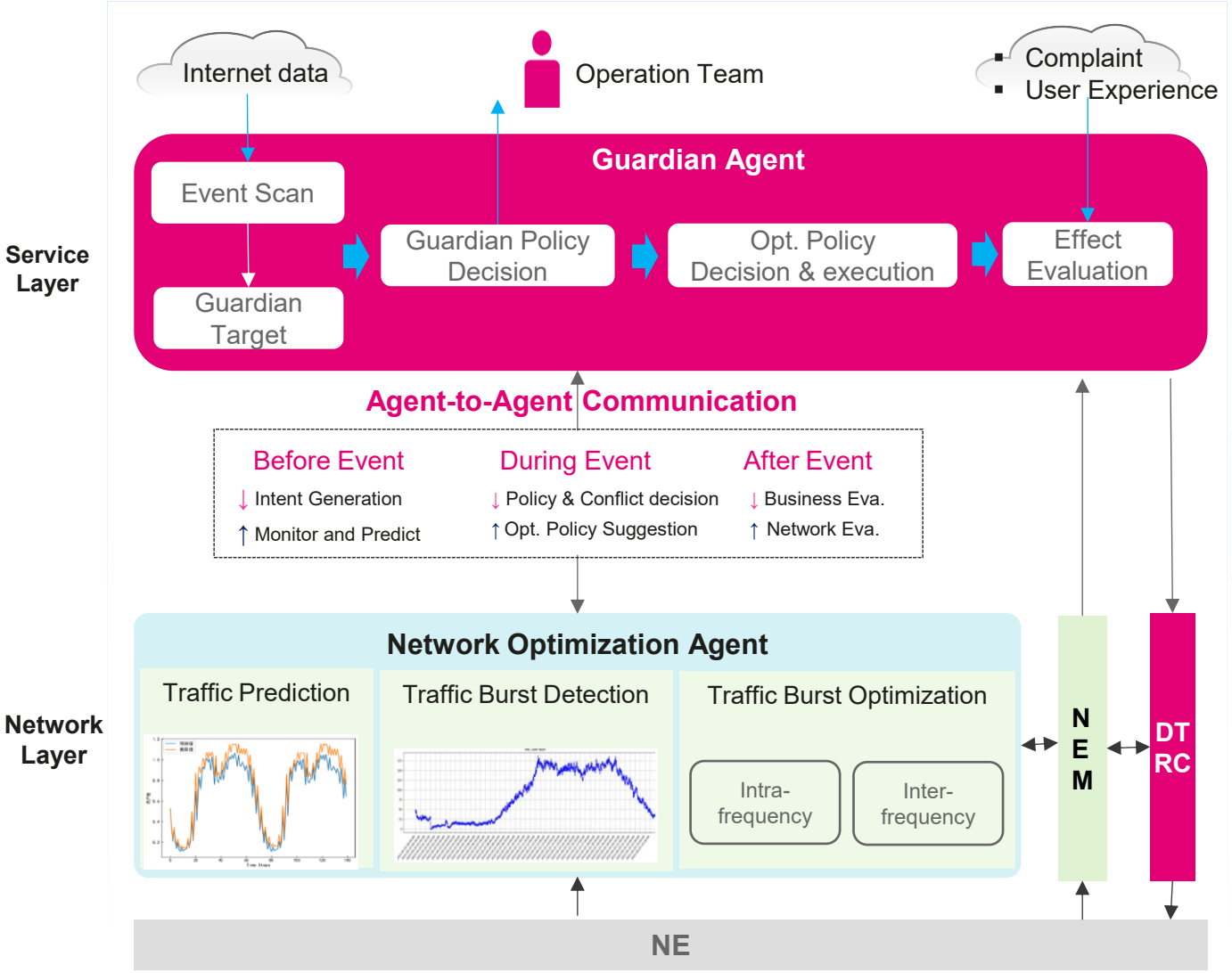
DT Agentic Target Architecture

- **Introducing agents at both service layer and network layer:**
  Multi-vendor and multi-domain autonomy enabled by single domain autonomy





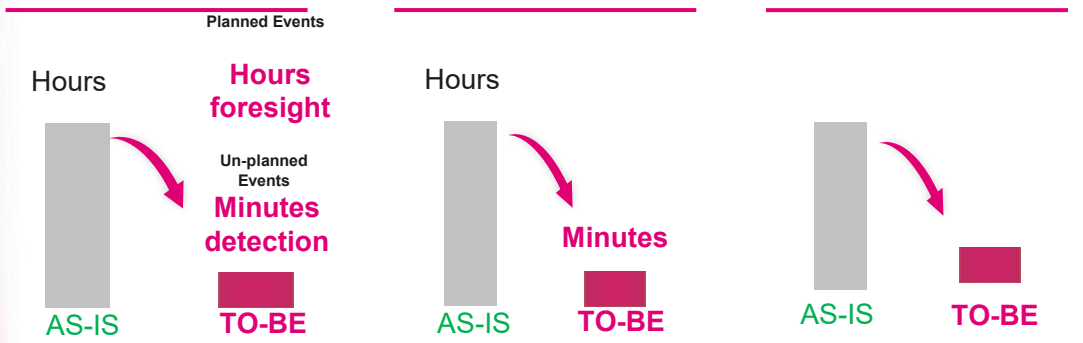
Service Experience Promotion enabled by Collaboration of Multi-layer agents



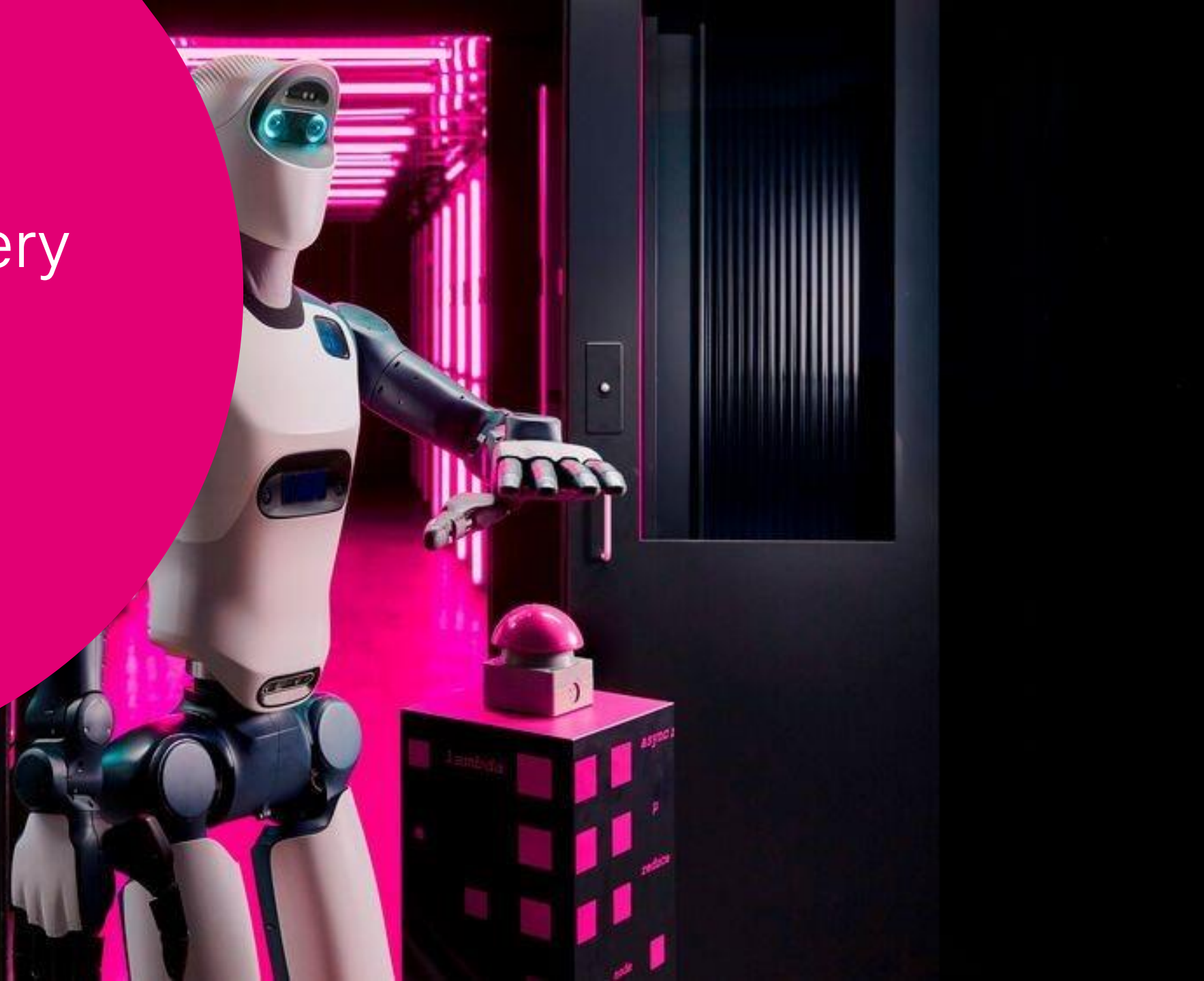
Scenarios

- **Planned Events**: 1500+ Events in Germany per year. for example, Beer Exchange, Concerts, Sports in the venue.
- **Un-planned Events :** traffic-jam, cell out-of-service.

Values



02
AI Agent Discovery



AI Discovery Motivation

AI Agents play a crucial role in modern telecommunications by enabling intelligent automation, decision-making, and adaptive network management.

These agents are software-driven entities that leverage artificial intelligence, including machine learning and natural language processing, to interact with users, applications, and network components.

Organisations require robust automatable mechanisms for the secure advertisement and discovery of AI agents on public and private networks.

- AI Agents need to be discoverable and resolvable
- Requirement to discover AI Agents in a multi-vendor environment
- Requirement to have a global stable unique identity across organisations and agreed governance
- Requirements for secure and trust resolution
- Prerequisite for flexible Agent-to-Agent/Tool/Entity communication

AI Discovery Current State

"Do something new"

- Add auth, security, etc.

"Handle it via the network"

- DNS, SIP-like registration with carrier networks, etc.

"Handle it via applications"

- Model cards & well-known URL

AI Discovery Current State

"Do something new" -> What happens if the IETF takes too long?

- Add auth, security, etc.

"Handle it via the network" -> What happens if the metadata is too rich?

- DNS, SIP-like registration with carrier networks, etc.

"Handle it via applications" -> What if application owners create dysfunction?

- Model cards & well-known URL

Working Draft

Versions: [draft-mozleywilliams-dnsop-dnsaid-01 - DNS for AI Discovery]

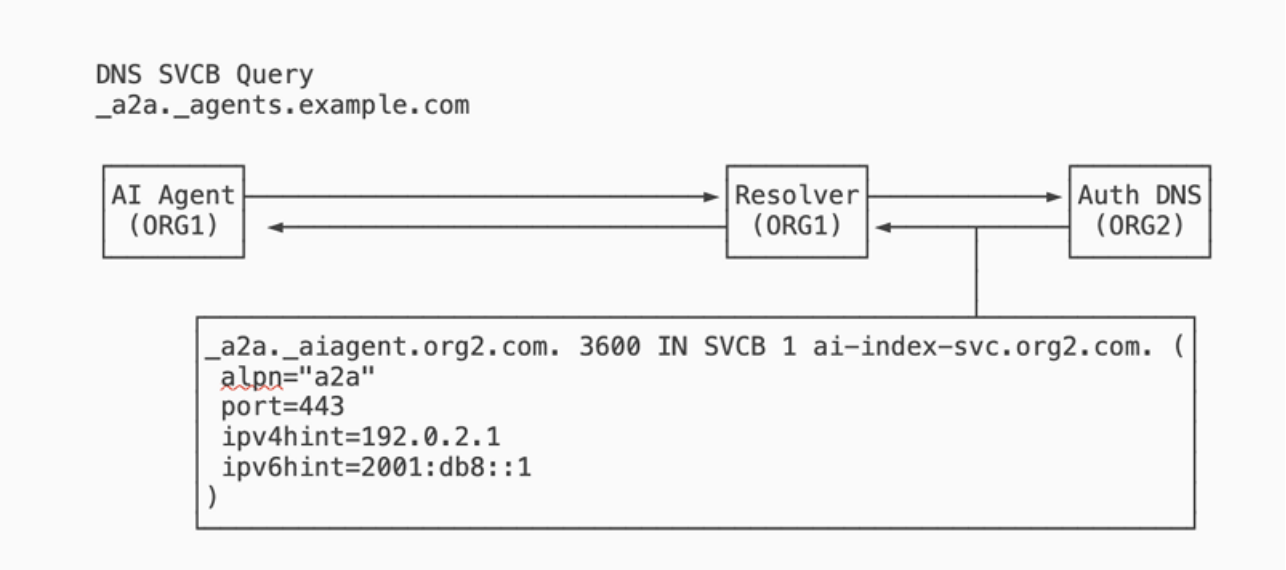
- **Guidance provided to operators to prevent DNS abuse (spam, record duplication, zones etc.)**

- **Metadata (metrics required in discovery)**
  - JSON model card URI, cost, location, modalities, etc. (yet to be standardized)

IANA to reserve a leaf attribute zone (e.g. _agents.example.com), SVCB params (e.g. cost, location, attestations, JSON model card URI, etc.) which need to be standardized by IETF WGs over time.

DNS-AID Example

In Progress



- SVCB params to be modified to include application-specific context
- (input token amounts, cost per input bundle, etc.)

Backup

AI Discovery Motivation Additions

Facilitating agent to agent discovery in a decentralized manner that won't lead to internet fragmentation, nor sovereignty issues with equal opportunity for ownership between circuit/network and app/developer personas.

Democratizing the use and access of AI solutions, disadvantaging walled garden approaches that lock consumers into proprietary registries. Delivered as a known entry point to an organization (e.g. index._agents.example.com), also opportunities for individual agent records.

- The contents / transactions between an index and agents, standardization efforts of model cards, etc. are reserved for AI working group activity, but this discovery method is an integration source for *any* protocol development (TBD)

Bonus: using existing technologies for rapid adoption.

- Create known subdomains (_agents.example.com) where SVCB records advertise agents (chat._mcp._example.com) and their capabilities through parameters (e.g. JSON model card's URI, cost, location, etc).
- Capability for index servers of all agents at example.com.
- Capability for registry services of all agents or indexes of agents at example.com and example2.com.

Cons: potentially imperfect discovery (some use cases excluded).